



OPEN A wastewater balance reveals unexpectedly high TFA input from human water use

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Trifluoroacetate (TFA) is a member of the PFAS group and shares their persistence. Its very small molecular size and anionic character makes it, in contrast to many other PFAS, highly water-soluble and mobile in the aquatic environment. Wastewater shows elevated TFA concentrations. This TFA may stem from background sources such as drinking water, infiltrating groundwater, or rainwater. It may also come from human water use itself. To quantify a potential TFA surplus, we built a mass balance at three rural sewer sites in Freiburg, Germany. End-member mixing analysis yielded TFA loads from groundwater and drinking water. Flow comparisons between dry and rain-event conditions yielded the TFA load from rainwater. Groundwater and drinking water contributed the largest background TFA load at two sanitary sewers. During a storm event rainwater dominated the TFA load in the combined sewer despite diluting its concentration. At every site, human water use exceeded the sum of the three background sources and made up the largest share of the TFA load. TFA concentrations also peaked during the morning wastewater surge at every site. We attribute this surplus mainly to human excretion of ingested TFA from food and drink. However, a comparison with an approximate dietary TFA intake estimate shows that this source alone cannot explain the observed surplus. Other sources, such as cosmetic products, may contribute to the overall TFA load in the sewer. Our study highlights the need for more data on human TFA uptake and excretion, as well as on other potential sources of TFA in human water use.

Keywords Trifluoroacetate (TFA), Sewer, Per- and polyfluoroalkyl substances (PFAS), Wastewater, Water stable isotopes, End-member mixing analysis (EMMA)

Trifluoroacetic acid is the smallest acid of the per- and polyfluoroalkyl substances (PFAS). Like other PFAS, it is chemically stable and environmentally persistent. Unlike longer-chain PFAS, however, trifluoroacetate (TFA, $pK_a = 0.23$) has a very small molecular size and a low distribution coefficient ($\log D \approx -2.62$)¹. This combination makes TFA exceptionally mobile in the environment, including soils and sediments, and also facilitates its use as a hydrological tracer². Together, persistence and mobility qualify TFA as a very persistent, very mobile (vPvM) substance³. Humans are exposed to TFA via multiple pathways.

TFA occurs in drinking water (DW) from several sources. Rising background concentrations in rainfall, which stem from the atmospheric degradation of fluorinated gases, partly contribute to this^{4–6}. Terrestrial TFA precursor compounds, particularly fluorinated pesticides applied in agriculture, cause further drinking water contamination when groundwater (GW) is used as drinking water source^{7–11}. Industrial and municipal wastewater treatment plants (WWTPs) also release TFA into rivers, which can reach drinking water abstracted from riverbank groundwater, or contaminate drinking water directly where surface water is the main abstraction source⁸. Drinking water regulatory limits vary within the European Union (EU) and largely depend on the data used for toxicological risk assessment (Netherlands: $2.2 \mu\text{g L}^{-1}$ ¹², Denmark: $9 \mu\text{g L}^{-1}$ ¹³, Germany: $60 \mu\text{g L}^{-1}$ ¹⁴). Changes in order to include TFA into the EU Drinking Water Directive, which currently does not specify a limit for TFA, were recently proposed^{15,16}.

In addition to drinking water, humans take up TFA via the consumption of foodstuffs and drinks. Plants accumulate TFA from soil water via evapotranspiration, leading to elevated concentrations in plant products such as fruits, vegetables, and grains^{17–21}. Again, the use of fluorinated pesticides may lead to extra TFA uptake. The processing of foods can lead to a further increase in TFA levels, as observed in products such as tea, beer, and

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wine^{22–25}. Furthermore, human exposure to TFA can also originate from medication. Within the human body, pharmaceuticals containing carbon-bound $-CF_3$ groups can partially degrade into TFA during metabolism^{8,9,26}. A recent screening identified 111 such per- and polyfluorinated active pharmaceutical ingredients approved for human use, 84% of which are potential TFA precursors²⁷. Once consumed or formed via metabolism, TFA is excreted through the kidneys and enters the wastewater (WW) system²⁸. Concentrations between 0.57 and 1.3 $\mu\text{g L}^{-1}$ were reported in influents of German WWTPs, 8 years ago⁸. A more recent survey of TFA in European groundwaters shows concentrations that are equal to or exceed this range¹¹. Groundwater is the dominant source of drinking water in Europe, and drinking water is the main component of wastewater. It remains unclear how much TFA in the sewer stems from the background concentration already present in drinking water, and how much is added through human water use itself, via blackwater (human excretion) or greywater (other household water use). We hypothesize that the TFA in groundwater, drinking water and rainwater cannot explain the total TFA load in the sewer system. To test this, we established TFA mass balances for three sewer catchments in south-west Germany, covering both a combined and two sanitary systems and contrasting land-use profiles, none of which contains known industrial TFA point sources. We separated the wastewater into its major components, drinking water, infiltrating groundwater and rainwater (RW), using water stable isotopes, and quantified the TFA load carried by each. Comparing the sum of these background loads with the total load measured in the sewer allows us to quantify a potential surplus. The origin of such a surplus, and in particular the role of human water use, is then addressed in the discussion.

Methods

Study sites

Our measurement sites were located in rural outskirts within the sewer network of the city of Freiburg, south-west Germany (Fig. 1), each representing a different land-use profile. The Pfaffenweiler catchment (site A) is drained by a combined sewer system, lies in a vineyard region and has 2630 population equivalents (PE). The Umkirch catchment (site B) is connected to a separated sewer system, includes a large commercial zone (38% of the area), a residential area with 5664 PE, and is surrounded by agricultural land. The Oberried catchment (site C) is also drained by a separated sewer system, includes several villages with 2352 PE, and is located within the forested mountainous terrain of the Black Forest. None of the catchments includes potential TFA point sources, such as fluoro-chemistry, pharma, or biotech industry. Our wastewater mass balance quantified the TFA loads of drinking water, groundwater, and rainwater at all three sites and, most importantly, a potential TFA surplus attributable to human water use.

Sampling

The sewer sampling was conducted from the 21st at 13:00 until the 23rd 11:00 in October 2024. Every two hours, a sample was taken by an ISCO 6712 Portable Sampler (Teledyne, Lincoln, NE, USA) into a fabric new 500 ml polyethylene bottle. Samples were stored on ice inside the sampler. The sampling points were equipped with a self-made precipitation sampler, which consisted of a polyethylene funnel, a polyethylene canister, and a ping-pong ball in the funnel for evaporation protection. From the sampler's bottles and precipitation canister, samples were taken in parallel in 100 ml brown glass bottles for later stable isotope analysis and ion chromatography of major ions, and fabric-new 50 ml polypropylene centrifuge tubes (Greiner, Kremsmünster, Austria) for

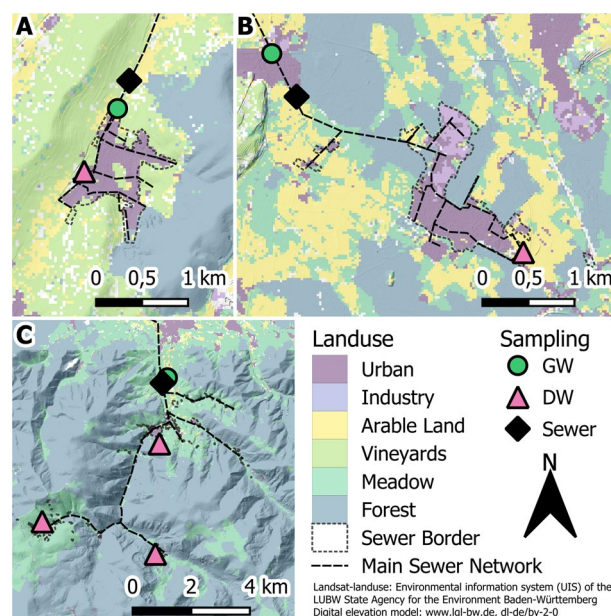


Fig. 1. Sewer catchments: land-use and sampling positions. Map created in QGIS 3.40 (<https://www.qgis.org/>); data sources and licence are credited on the map.

subsequent LC-MS TFA analysis. Samples were stored in the refrigerator at 4 °C for a maximum of 2 weeks before analysis. We measured conductivity using a handheld meter (LF325 WTW, Weilheim, Germany) before redistributing the samples for further laboratory analysis. On the 22th at 09:00 the sampler at site A was clogged and did not draw a sample. Linear interpolation of TFA and isotope values filled the resulting gap, allowing us to perform the end-member mixing analysis (EMMA) and flow separation despite the missing sample.

We sampled groundwater once at each site, by pumping from the sampling well until the electric conductivity remained constant. Afterwards, a sample was taken in the vials mentioned above (brown glass bottles, polypropylene tubes). Catchments A and B are small, and suitable groundwater wells were scarce. Because no sampling well was available at site A, we drilled a borehole half a meter below the groundwater level next to the sewer. We took the sample directly from the borehole without further pumping. This was done to prevent drawing bank filtration water from the stream Duffernbach, located 1 m next to the borehole. At C, the larger catchment, we sampled groundwater at the catchment outlet. This integrates the signal over the whole catchment, so we considered a single sample representative there too.

Drinking water samples were taken from the tap after 3 min of the faucet being open. Drinking water at sites A and B each come from a single source: at A, it is imported from outside the catchment with no local springs; at B, all of it comes from the same well. We therefore expected both to be spatially uniform and sampled them once. Drinking water at site C has different sources for each subarea of the catchment. We sampled drinking water from the supply wells at St. Wilhelm, Hofgrund, and the village of Oberried. We weighted the found concentrations by the number of inhabitants (260, 580, 1992) to calculate the mean.

High-pressure liquid chromatography-tandem mass spectrometry (HPLC MS/MS) TFA analysis

Samples were transferred into 1.5 mL Eppendorf tubes and centrifuged for 15 min at 18,000g (Centrifuge 5417, Eppendorf, Germany). A 1 mL aliquot was transferred into LC vials (Isera Düren) and moved to the pre-cooled (4 °C) autosampler. Ion exchange liquid chromatography-electrospray tandem mass spectrometry (LC-ESI MS/MS) analysis using a Shimadzu LC-AD20 coupled with API 5500 Q Trap triple-quadrupole mass spectrometer (Applied Biosystems/MDS Sciex Instruments, Concord, ON, Canada) with an electrospray interface operating in negative ionization mode was applied for the detection of TFA. The injection volume was set to 50 µL, with separation performed on an IonPac AS17-C column (2 × 250 mm) and an IonPac AG17-C guard column (2 × 50 mm) (both from Thermo Fisher Scientific, Waltham, USA). The column compartment was kept at 40 °C and a flow rate of 0.28 mL min⁻¹. The method is based on the method described by Scheurer et al.⁸. The analysis was conducted using an ultra-pure water solution containing 50 mM ammonium hydrogen carbonate (Honeywell Fluka™) (A) and methanol (ROTISOLV® 99.98 %, Ultra LC-MS Grade Carl Roth) (B) as eluents. The used columns were pre-equilibrated with the starting conditions of 20% A, and all samples were analyzed using the following gradient: 20% of eluent A (0–1 min), 20–50% A (1–10 min), 50–20% A (11–16 min). Calibration was performed using a precisely prepared dilution series of TFA standards (pure trifluoroacetic acid, obtained from Sigma-Aldrich) in ultra-pure water. We spiked each sample with 1 µg L⁻¹ TFA-¹³C₂ (LGC Standards) as internal standard (IS), adding 1% of a 100 µg L⁻¹ IS solution to each sample. UltraPure H₂O (18.2 MΩ cm) was prepared in-house with a Milli-Q® Direct Water Purification System (Thermo-scientific®). MS parameters for both compounds are listed in the Supplement to this article (Table S2). Samples were measured in duplicate from the same vial and averaged, with a mean relative standard deviation of 1.7% over all duplicates, calculated as described in²⁹. A limit of quantification (LOQ) of 0.08 µg L⁻¹ was determined according to DIN 32645 from the standard deviation (SD) of blank values (limit of detection, LOD = 10 × SD). Because contamination is a concern, all used laboratory and field equipment (for example, the precipitation measurement setup, ISCO bottles, Falcon tubes, Tips, and Eppendorf tubes) was washed with UltraPure H₂O and/or Ultrapure MeOH and analyzed by the method described above. None of the equipment used showed relevant background levels of TFA. Consequently, blank correction of the analytical results was not required. To ensure a contamination-free HPLC MS/MS system, we sampled Milli-Q® until background levels were below 20 ng L⁻¹, which we considered negligible.

Ion chromatography (IC) major ions and cavity ring-down spectroscopy (CRDS) isotope analysis

From the 100 mL glass bottles, 5 mL aliquots were filtered using 0.45 µm syringe Millipore filters (VWR, Darmstadt, Germany) and analyzed for major anions (chloride, nitrate, sulfate) and cations (sodium, potassium, magnesium, calcium, ammonium) via IC (DIONEX ICS-1100, Thermo Fisher Scientific, Waltham, USA). For separation, we used an IonPac AS17-C column (4 × 250 mm) and an IonPac AG17-C guard column (4 × 50 mm) (both from Thermo Fisher Scientific, Waltham, USA). Thermo Fisher Scientific reported a precision of 5% and an LOQ of 1 mg L⁻¹ for the IC system. Only nitrate and ammonium were used in the analysis. The remaining ions were obtained within the same chromatographic run and are reported in the data supplement for transparency and to enable reuse of the dataset. Additionally, 1 mL aliquots were analyzed for stable isotopes using CRDS with a L2130-*i* System (Picarro, Santa Clara, USA), achieving supplier-defined precisions of 0.16‰ for δ¹⁸O and 0.6‰ for δ²H.

EMMA and separation of rainwater

To estimate the amount of infiltrating groundwater, we used water stable isotopes (δ²H and δ¹⁸O), a method successfully applied in previous studies^{30,31}. The method assumes that domestic drinking water use does not significantly alter its isotopic composition. Water that has passed through human metabolism can be altered isotopically, but this shift becomes negligible once diluted by the much larger drinking water and groundwater volumes in wastewater³². However, high-efficiency condensing boilers can affect the isotopic composition of wastewater. In these systems, the exhaust gas cools below 100 °C, causing combustion water to condense and be

discharged into the sewer. Its oxygen originates from atmospheric O_2 and can therefore alter the isotopic oxygen signal of domestic wastewater despite the relatively small input volumes. In contrast, the hydrogen originates from fossil organic matter and has an isotopic composition more similar to that of the hydrogen already present in natural waters³². We therefore used the less affected δ^2H as a tracer to separate groundwater from drinking water by linear EMMA.

We separated the measurement period into two days. It was dry on the first day (2024-10-21 13:00–2024-10-22 11:00), but it rained on the second (2024-10-22 13:00–2024-10-23 11:00). The first day was used to separate the groundwater fraction by EMMA and the second to separate the rainwater fraction by flow-difference. All sewers lie in groundwater, hence we used groundwater and drinking water as end members. Throughout this and the following section, σ denotes the propagated standard error of the respective quantity. These errors originate from the relative standard deviation of duplicate TFA measurements (1.7%, see HPLC MS/MS section above) and from the supplier-stated precision of the CRDS isotope analysis (0.16‰ for $\delta^{18}O$, 0.6‰ for δ^2H , see CRDS section above). The flow of groundwater in the sewer (Q_{GW} , $L\ s^{-1}$) was calculated using deuterium, measured in wastewater (δ_{WW} , ‰) and in drinking water and groundwater (δ_{DW} , δ_{GW} , ‰) by Eq. 1.

$$Q_{GW}(t) = X(t) \times Q_{WW} \quad (1)$$

The flow of drinking water (Q_{DW} , $L\ s^{-1}$) is then calculated according to Eq. 2, with $X(t)$ being the proportion of groundwater, as defined in Eq. 3.

$$Q_{DW}(t) = (1 - X(t)) \times Q_{WW} \quad (2)$$

$$X(t) = \frac{\delta_{WW} - \delta_{DW}}{\delta_{GW} - \delta_{DW}} \quad (3)$$

The error of the groundwater ($\sigma_{Q_{GW}}(t)$, $L\ s^{-1}$) and drinking water ($\sigma_{Q_{DW}}(t)$, $L\ s^{-1}$) flow propagates from the error in the discharge measurement ($\sigma_{Q_{WW}}$, $L\ s^{-1}$, provided by the Abwasserzweckverband (AZV) Breisgauer Bucht, the local wastewater treatment association) and the error in the groundwater fraction ($\sigma_{X(t)}$, $L\ s^{-1}$) as shown in Eqs. 4 and 5.

$$\sigma_{Q_{GW}}(t) = \sqrt{[X(t) \times \sigma_{Q_{WW}}]^2 + [Q_{WW} \times \sigma_{X(t)}]^2} \quad (4)$$

$$\sigma_{Q_{DW}}(t) = \sqrt{[(1 - X(t)) \times \sigma_{Q_{WW}}]^2 + [Q_{WW} \times \sigma_{X(t)}]^2} \quad (5)$$

An error estimation for $\sigma_{X(t)}$ can be found in the literature³⁰. We calculated the standard error of the resulting daily mean groundwater fraction from the standard error of the mean, as shown in Eq. 6, where n is the number of samples in a day.

$$\sigma_{\bar{f}} = \frac{\sigma_f}{\sqrt{n}} \quad (6)$$

To obtain the amount of infiltrating groundwater on day two, we assumed that the absolute amount at a given time on day one is equal to the amount at the same time on day two. The amount of rainwater was calculated by comparing the discharge on the dry day ($Q_{d1}(t)$, $L\ s^{-1}$) with the discharge on the rainy day ($Q_{d2}(t)$, $L\ s^{-1}$) as shown in Eq. 7.

$$Q_{RW}(t) = Q_{d2}(t) - Q_{d1}(t) \quad (7)$$

The error of the absolute rainwater flow ($\sigma_{Q_{RW}}(t)$, $L\ s^{-1}$) is calculated from the error of the rainwater fraction ($\sigma_{X(t)}$, dimensionless) and the error in the discharge measurements $\sigma_{Q_d}(t)$ as described in Eq. 8.

$$\sigma_{Q_{RW}}(t) = \sqrt{[\sigma_{Q_{d2}}(t)]^2 + [\sigma_{Q_{d1}}(t)]^2} \quad (8)$$

Calculating loads

Loads ($L_i(t)$, $\mu g\ s^{-1}$) were calculated by multiplying the flow (Q_i , $L\ s^{-1}$) and TFA concentration ($c_i(t)$, $\mu g\ L^{-1}$) in groundwater, drinking water and rainwater (Eq. 9).

$$L_i(t) = Q_i(t) \times c_i(t) \quad (9)$$

The loads error ($\sigma_{L_i}(t)$, $\mu g\ s^{-1}$) is propagated from the error in the flow ($\sigma_{Q_i}(t)$, $L\ s^{-1}$) and the error in the concentration ($\sigma_{c_i}(t)$, $\mu g\ L^{-1}$) (Eq. 10).

$$\sigma_{L_i}(t) = \sqrt{[Q_i(t) \times \sigma_{c_i}(t)]^2 + [c_i(t) \times \sigma_{Q_i}(t)]^2} \quad (10)$$

We calculated the combined background load $L_C(t)$ (Eq. 11) and its error ($\sigma_{L_C}(t)$, $\mu g\ s^{-1}$) according to Eq. 12.

$$L_C(t) = L_{DW}(t) + L_{GW}(t) + L_{RW}(t) \quad (11)$$

$$\sigma_{L_C}(t) = \sqrt{[\sigma_{L_{DW}}(t)]^2 + [\sigma_{L_{GW}}(t)]^2 + [\sigma_{L_{RW}}(t)]^2} \quad (12)$$

The actual loads in the sewer system ($L(t)$, $\mu\text{g s}^{-1}$) calculated from the measured TFA concentration in the sewer system ($c(t)$, $\mu\text{g L}^{-1}$) and the total discharge ($Q_{WW}(t)$, L s^{-1}) as (Eq. 13).

$$L(t) = Q_{WW}(t) \times c(t) \quad (13)$$

The AZV measured discharge by magnetic inductive flow meters (MID) at site A, B, and C. The measurement error was estimated as the error to a second portable MID at A, B, and C (1.4, 4.0, and 4.7%). MID's were flushed before sampling. The error of the actual load ($\sigma_{L(t)}$, $\mu\text{g s}^{-1}$) is calculated from the error in the flow ($\sigma_{Q_{WW}}(t)$, L s^{-1}) and the error in the concentration ($\sigma_c(t)$, $\mu\text{g L}^{-1}$) (Eq. 14).

$$\sigma_L(t) = \sqrt{(c(t) \times \sigma_{Q_{WW}}(t))^2 + (Q_{WW}(t) \times \sigma_c(t))^2} \quad (14)$$

To ensure comparability, we normalized the difference between measured loads in the sewer and the ones calculated from the wastewater components ($\Delta L(t)$, $\mu\text{g s}^{-1}$) by the total number of connected inhabitants per sewer catchment (N ; 2630, 5664, and 2352 at A, B, and C, respectively, from the Statistical Office of Baden-Württemberg) as shown in Eq. 15 with the corresponding error ($\sigma_{L(t)norm}$, $\mu\text{g s}^{-1}$) (Eq. 16).

$$L(t)_{norm} = \frac{L(t) - L_C(t)}{N} \quad (15)$$

$$\sigma_{L(t)norm} = \frac{\sqrt{\sigma_{L(t)}^2 + \sigma_{L_C(t)}^2}}{N} \quad (16)$$

We calculated $L(t)$ and $L(t)_{norm}$ and the two-day means, and multiplied by the number of seconds per day to obtain daily loads. The mean error was calculated according to Eq. 17 where n is the number of samples (24), c is the number of seconds per day, and $L(t)$ is a load calculated at one of the sampling times. σ_{Lnorm} was calculated in analog to σ_L .

$$\sigma_L = c \times \frac{\sum \sigma_{L(t)}^2}{n} \quad (17)$$

Results

Wastewater flow exhibited a typical diurnal pattern, which was comparable to dry conditions from the last seven years (see Supplement Figure S1). Sewer TFA concentrations exhibited similar diurnal variation and remained above the drinking water and groundwater end members at all sites (Fig. 2). The vineyard region A showed the highest overall concentrations (mean: $1.27 \pm 0.22 \mu\text{g L}^{-1}$), while the predominantly forested catchment C had

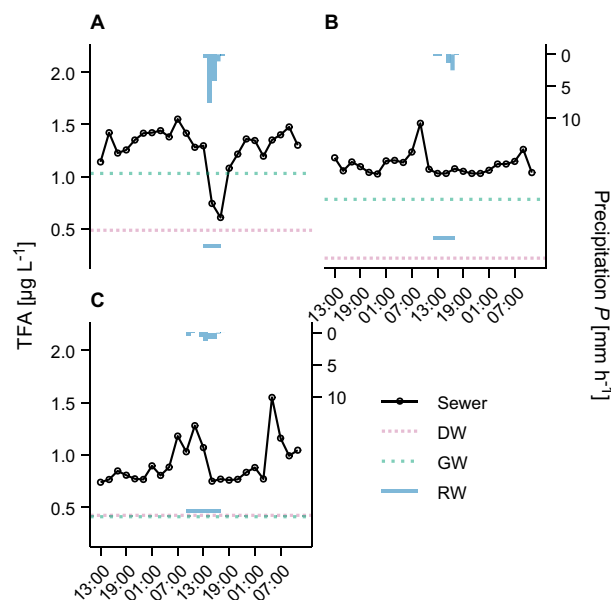


Fig. 2. TFA concentrations in the sewer system, in the end members drinking water (DW) and groundwater (GW), and in precipitation, i.e. rainwater (RW). Panels A, B, and C correspond to the three study sites. Top-down bars indicate precipitation volumes at the nearest climate station (distance to sampling spot < 7.5 km).

lower concentrations (mean: $0.92 \pm 0.21 \mu\text{g L}^{-1}$). All stations displayed a distinct morning TFA peak between 5:00 and 11:00 AM, accompanied by a rise in ammonium (NH_4^+) concentrations (see Supplement Figure S2). Following the rainfall event during the second day of sampling, TFA concentrations at A decreased and nearly matched those of rainfall. In contrast, TFA in the sanitary sewers at B and C remained relatively stable. During the first sampling day, EMMA with $\delta^2\text{H}$ yielded rather constant dry-weather groundwater contributions with a slight diurnal pattern (see Supplement Figure S3 for mixing lines). These were comparable at sites A ($59 \pm 7\%$) and B ($49 \pm 4\%$), within their combined uncertainty, and lower at C ($40 \pm 5\%$). The absolute amounts of groundwater and drinking water followed the typical daily pattern of wastewater emergence, they were minimal during the night and peaked in the early morning (Fig. 3). The rainfall event, measured by our precipitation samplers, totaled 10.9 mm at A, 4.2 mm at B, and 3.8 mm at C. This event during the second day of sampling only slightly raised discharge in the sanitary sewers of sites B and C, while we observed a sharp peak in the combined sewer of A. Subtracting dry-weather flow rates of the previous day, remaining rainwater fractions were marginal at the sewers B and C (up to 8%, 19%), but considerable at A (up to 91%), where they formed a distinct peak and increased wastewater flow during the entire second day. This prolonged rainwater contribution was also reflected in enriched $\delta^2\text{H}$ values (see Supplement Figure S4). When we multiplied drinking water and groundwater components with their respective end member concentrations, the resulting TFA loads of these components followed the typical diurnal wastewater cycle, with higher TFA transport during daytime and lower values at night (Fig. 4). The same multiplication for rainwater yielded additional TFA inputs into the sewer systems, which fell below the standard errors in the sanitary sewers of B and C, but dominated the TFA balance in the combined sewer of A during rainfall. The TFA concentration in rainfall was low ($0.33 \mu\text{g L}^{-1}$), so rainwater dominated the load by its volume, not by its concentration. When we compared the sum of the estimated loads from drinking water, groundwater, and rainwater with the actually measured TFA loads in the sewers, we found the largest discrepancies during the early morning hours at all stations and at the onset of the rain event at region A. In general, actual TFA loads in the sewer system exceeded those calculated from drinking water, groundwater, and rainwater in all samples over the entire sampling period. This pointed to a major source of TFA from human water use, which contributed approximately half (44–55%) of the total TFA load (Fig. 5). When we used inhabitant numbers to break this surplus down on daily per capita TFA input, we arrived at $130\text{--}206 \mu\text{g d}^{-1} \text{capita}^{-1}$ (Table 1). After human water use, groundwater and drinking water were the next largest contributors to TFA. In regions B and A, groundwater accounted for a slightly larger share than drinking water (35% and 32%, respectively). This pattern was reversed in the steep, predominantly forested region C, where drinking water contributed more TFA than groundwater. Rainwater only made a substantial contribution in the mixed sewer of region A, where it accounted for 13% of the total TFA load across both sampling days.

Discussion

Our approach disclosed overall patterns of TFA input into the sewer systems. During rainfall, we observed temporarily high TFA loads in the combined sewer at A, although TFA concentrations dropped down to rainwater levels during the event. This drop reflects simple dilution by rainwater. However, the surplus load that is not explained by the background sources rainwater, drinking water and groundwater peaked at the same

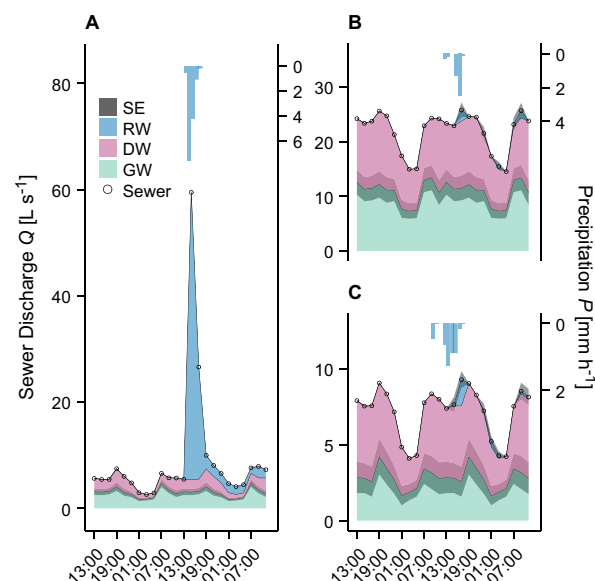


Fig. 3. Wastewater components drinking water (DW) and groundwater (GW) and their contribution to total flow, separated by $\delta^2\text{H}$ end-member mixing analysis (EMMA). Precipitation is plotted from the top, resulting in rainwater (RW), calculated by the difference in flow. Panels A, B, and C correspond to the three study sites. Error bands are marked in grey and are given as standard errors (SE).

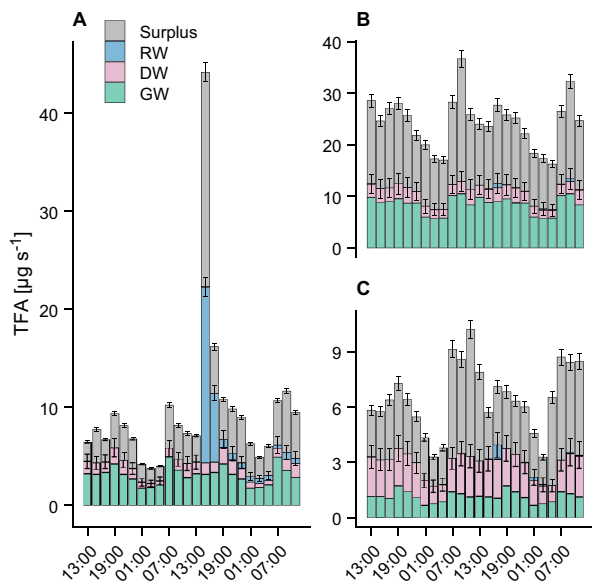


Fig. 4. TFA loads in the sewer, stacked by source: groundwater (GW), drinking water (DW), and rainwater (RW). The grey segment shows the surplus of the total measured wastewater (WW) load that is not explained by these three sources (WW-RW-DW-GW). Error bars indicate standard errors (SE) on the combined GW+DW+RW subtotal and on the total measured WW load. Panels A, B, C correspond to the three study sites.

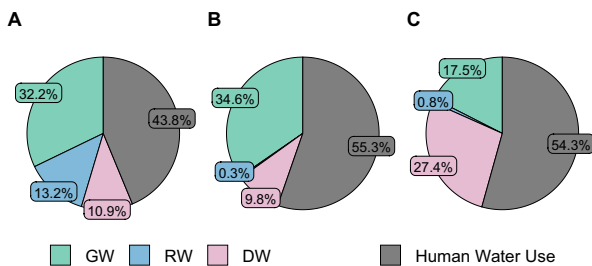


Fig. 5. Shares on the total TFA loads in the sewer over the whole sampling period of two days. The difference between values from groundwater (GW), drinking water (DW), and rainwater (RW) and the total load measured in the sewer, which we attributed to human water use, is marked in grey. Panels A, B, C correspond to the three study sites.

Station	TFA load (measured, WW) [$\mu\text{g d}^{-1}$]	Surplus (human water use) [$\mu\text{g d}^{-1}$]	Surplus fraction	Surplus per capita [$\mu\text{g d}^{-1} \text{ capita}^{-1}$]
A	824 ± 5	361 ± 14	44 ± 2	137 ± 4
B	2105 ± 19	1165 ± 25	55 ± 1	206 ± 6
C	564 ± 6	306 ± 13	54 ± 2	130 ± 4

Table 1. Daily TFA loads and the surplus attributable to human water use, i.e. the difference between the measured wastewater (WW) load and the sum of the drinking water (DW), groundwater (GW), and rainwater (RW) loads.

time (Fig. 4). This points to a mechanism adding TFA to the sewer beyond these three sources. One possible mechanism is mobilization within the sewer. TFA might sorb to organic soil, as shown in laboratory experiments at high concentrations³³, and is taken up by microorganisms³⁴. Sewer biofilms and sludge consist mainly of microorganisms and organic material. Temporarily stored TFA could mobilize during high flow, a phenomenon previously observed for other micropollutants^{35,36}. A second possible mechanism acts on the way to the sewer. Rainwater can pick up additional TFA before it arrives. First, dry-deposited TFA may accumulate on surfaces between rain events³⁷, and runoff can wash this off into the sewer (first flush effect³⁸). Our precipitation sampler

also collects some dry deposition, but its smooth funnel likely retains less than rough surfaces like streets or grass, so it might underestimate this wash-off contribution. Second, at site A, rainwater travels down the sloping vineyards onto the streets, where it may pick up TFA through soil leaching. Taking the low log D of TFA into account, we think that TFA accumulation during runoff generation is likely the dominant mechanism.

During dry conditions, the result of our mixing analysis depended on end member concentrations and characteristics of the contributing catchments and sewer networks. If TFA concentrations are lower in groundwater than in drinking water, drinking water is more significant. The region C example illustrates this. TFA from groundwater is more relevant where groundwater TFA concentrations markedly exceed the precipitation background of $0.335 \mu\text{g L}^{-1}$ ¹⁶. We found this in the agricultural A and B catchments. The agricultural TFA might originate from pesticide usage^{11,39,40} or from liquid manure⁴¹. This attribution remains speculative with our data.

Volumes of extraneous groundwater infiltration may vary and depend on the technical condition of the sewer network and on its relative position to surrounding groundwater. In general, our dry weather groundwater contributions compare to those found in other places⁴². Earlier research has suggested that diurnal patterns of groundwater intrusion arise from the accumulation of backwater zone volumes in the sewer system³⁰. In every single sample we found a surplus of TFA that we attribute to human water use, amounting to $130\text{--}206 \mu\text{g d}^{-1}$ per capita (Table 1). The surplus followed the typical diurnal cycle of urban wastewater discharge, with lows during the night and peaks in the morning. Human water use peaks in the early morning hours, primarily due to toilet flushing, showering and washbasin drainage^{43,44}, so the elevated morning loads point to human excretion or to personal care products used during the morning routine. The parallel spiking of TFA and NH_4^+ concentrations specifically points to urine.

TFA is present in various foodstuffs^{18–20,22,24} and is excreted mainly via the kidneys^{28,45}, so dietary intake should translate into a comparable urinary load. We therefore estimated daily dietary TFA intake independently, combining literature concentrations in food and beverages with German consumption statistics⁴⁶ (see Supplement Table S1). This estimate averaged $50.9 \mu\text{g d}^{-1}$ per adult ($52.1 \mu\text{g d}^{-1}$ for men, $49.7 \mu\text{g d}^{-1}$ for women).

Reported urinary TFA levels are of the same order. To make them comparable to our loads, we converted the published concentrations to daily excretion assuming an average urine output of 1.4 L d^{-1} per capita⁴⁷. For US citizens, one study found a detection frequency of 31% ($n = 81$; detection limit $3.5 \mu\text{g L}^{-1}$, equivalent to $5 \mu\text{g d}^{-1}$)⁴⁸; reanalysis showed that most individuals excreted between 5 and $17 \mu\text{g d}^{-1}$ ($n = 22$), while three individuals reached $78\text{--}406 \mu\text{g d}^{-1}$. Pooled urine samples from Australian citizens ($n = 70$ pools from 6040 individuals) yielded higher values of $5\text{--}420 \mu\text{g d}^{-1}$ with a median of $34 \mu\text{g d}^{-1}$ ⁴⁹. Our dietary estimate thus falls within the reported range and close to the Australian median, which supports the plausibility of both.

Dietary intake and subsequent urinary excretion are therefore the most likely single source of the surplus, but they are not sufficient on their own: our estimated $50.9 \mu\text{g d}^{-1}$ account for only about a third of the $130\text{--}206 \mu\text{g d}^{-1}$ observed. Our dietary estimate also carries substantial uncertainty and depends strongly on the assumed cereal and wine intake (see Supplement). Still, even the upper end of the reported excretion data does not close the gap. A substantial fraction of the surplus must therefore enter the sewer without passing through the body, which points to greywater and small household sources: cosmetic products contain TFA directly rather than only CF_3 precursors⁵⁰, and Teflon™ and Kel-F™ products emit TFA when heated⁵¹.

These interpretations rest on a limited observational basis. Each site was sampled only once, over 48 hours. Hence, the campaign cannot capture seasonal or year-to-year variability. Our conclusions on the role of rainwater are further constrained: sites B and C are drained by separate sewers, where little rainwater is expected by design, and the rainfall event was weak at both (4.2 and 3.8 mm) compared to the 10.9 mm recorded at the combined sewer of catchment A. Statements on the behaviour of separate sewers during larger rainfall events would require additional sampling.

Within these constraints, our balance consistently revealed more TFA in wastewater than the background sources rainwater, drinking water and groundwater can explain. We attribute a part of this surplus to human TFA uptake, and direct measurements in urine support this attribution, although knowledge of human TFA excretion remains scarce, particularly for the faecal route. Other sources likely add to the sewer load as well, such as cosmetic products containing TFA⁵⁰. Two challenges therefore remain: our estimate of human TFA uptake still carries substantial uncertainty, which better data on TFA in food and beverages could possibly reduce, and part of our identified TFA surplus still lacks an identified source. Since a full breakdown of every pathway would require considerable effort, a broader restriction of TFA precursor PFAS use may be the more practical response to the elevated TFA levels we observed.

Data availability

The data and code supporting the findings of this study are openly available at <https://doi.org/10.60493/b7gxxv-k y759>.

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Author contributions

Conceptualization IF, PH, JL; Methodology IF, PH, BH; Software IF, PH; Formal analysis IF, PH; Investigation IF, PH, BH, DN; Resources; JL, MM; Data Curation IF, PH; Writing—Original Draft, IF, JL; Writing—Review and Editing IF, PH, BH, DN, MM, JL; Visualization IF; Supervision JL, MM; Project administration JL, MM; Funding acquisition JL, MM.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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